Table of Contents

SmartUIScaler ---- 2
- CoreUIScaler ---- 3
- SmartUIManager ---- 6
- UIScalerAlignment ---- 7
- UIScalerBehaviour ---- 9
- UIScalerGroupSo ---- 14
- UIScalerGroups ---- 16
- UIScalerGroupsSaveData ---- 21
- UIScalerGroupsSaveData.UIScalerGroupOptions ---- 23
- UIScalerOptions ---- 25

Namespace SmartUIScaler

Classes

CoreUIScaler

Base class of UIScaler components with core scaling logic.

SmartUIManager

Entry point for external code that needs to stay in sync with UIScaler updates. Drives UIScaler updates in play mode and in edit mode. Created automatically and must not be added to a GameObject manually.

UIScalerBehaviour

Base class of every UIScaler related components.

UIScalerGroupSo

UIScaler group scriptable object.

UIScalerGroups

UIScaler groups manager.

UIScalerGroupsSaveData

UIScaler groups save data.

UIScalerGroupsSaveData.UIScalerGroupOptions

UIScaler group options save data.

Structs

UIScalerOptions

UIScaler options.

Enums

UIScalerAlignment

Used for aligning UIScaler inside provided screen area.

Class CoreUIScaler

Namespace: SmartUIScaler

Assembly: SmartUIScaler.dll

Base class of UIScaler components with core scaling logic.

```
public abstract class CoreUIScaler : UIScalerBehaviour
```

Inheritance

object ← Object ← Component ← Behaviour ← MonoBehaviour ← UIScalerBehaviour ← CoreUIScaler

Inherited Members

UIScalerBehaviour.RectTransform , UIScalerBehaviour.Awake() , UIScalerBehaviour.OnDestroy() , UIScalerBehaviour.OnTransformParentChanged() , UIScalerBehaviour.SetRectTransformDefaults() , UIScalerBehaviour.GetFirstParentCoreUIScaler() , UIScalerBehaviour.WorldRectToLocalRect(Transform, Rect) , UIScalerBehaviour.LocalToWorldPoints(Transform, Vector3, Vector3) , UIScalerBehaviour.WorldToLocalPoints(Transform, Vector3, Vector3) , UIScalerBehaviour.GetScreenSafeAreaRect()

Methods

GetScreenRect()

Called to receive the provided screen area rect.

```
protected abstract Rect GetScreenRect()
```

Returns

Rect

Provided screen area rect.

OnDisable()

```
protected override void OnDisable()
```

OnEnable()

```
protected override void OnEnable()
```

OnPreWillRenderCanvases()

Event that is called just before Canvas rendering happens.

```
protected override void OnPreWillRenderCanvases()
```

Reset()

```
protected virtual void Reset()
```

Resize()

Method where main resize logic should be implemented.

```
protected override void Resize()
```

ScreenRectToWorldRect(Rect)

Called to convert provided screen area rect into world rect.

```
protected abstract Rect ScreenRectToWorldRect(Rect screenRect)
```

Parameters

`screenRect` Rect

Screen area rect to be converted.

Returns

Rect

World space rect.

Class SmartUIManager

Namespace: SmartUIScaler

Assembly: SmartUIScaler.dll

Entry point for external code that needs to stay in sync with UIScaler updates. Drives UIScaler updates in play mode and in edit mode. Created automatically and must not be added to a GameObject manually.

```
[InitializeOnLoad]
[AddComponentMenu("")]
[DisallowMultipleComponent]
public class SmartUIManager : MonoBehaviour
```

Inheritance

object ← Object ← Component ← Behaviour ← MonoBehaviour ← SmartUIManager

Events

AfterScalersUpdatedEvent

Invoked every frame after UIScalers are updated. Read UIScaler sizes and positions here so external UI stays in sync within the same frame.

```
public static event Action AfterScalersUpdatedEvent
```

Event Type

Action

Enum UIScalerAlignment

Namespace: SmartUIScaler

Assembly: SmartUIScaler.dll

Used for aligning UIScaler inside provided screen area.

```
public enum UIScalerAlignment
```

Fields

`Bottom = 2`

Bottom middle of provided screen area.

`BottomLeft = 7`

Bottom left of provided screen area.

`BottomRight = 8`

Bottom right of provided screen area.

`Center = 0`

Center of provided screen area.

`Left = 3`

Middle left of provided screen area.

`Right = 4`

Middle right of provided screen area.

`Top = 1`

Top middle of provided screen area.

`TopLeft = 5`

Top left of provided screen area.

`TopRight = 6`

Top right of provided screen area.

Class UIScalerBehaviour

Namespace: SmartUIScaler

Assembly: SmartUIScaler.dll

Base class of every UIScaler related components.

```
[RequireComponent(typeof(RectTransform))]
public abstract class UIScalerBehaviour : MonoBehaviour
```

Inheritance

object ← Object ← Component ← Behaviour ← MonoBehaviour ← UIScalerBehaviour

Derived

CoreUIScaler

Fields

RectTransform

RectTransform of this behaviour

```
protected RectTransform RectTransform
```

Field Value

RectTransform

Methods

Awake()

```
protected virtual void Awake()
```

GetFirstParentCoreUIScaler()

Get the first indirect parent CoreUIScaler.

```
protected CoreUIScaler GetFirstParentCoreUIScaler()
```

Returns

CoreUIScaler

Indirect parent CoreUIScaler

GetScreenSafeAreaRect()

Get whole screen safe area.

```
protected static Rect GetScreenSafeAreaRect()
```

Returns

Rect

Whole screen safe area.

LocalToWorldPoints(Transform, Vector3, Vector3)

Converts local space points to world space.

```
protected static (Vector3, Vector3) LocalToWorldPoints(Transform localTransform, Vector3
minPoint, Vector3 maxPoint)
```

Parameters

`localTransform` Transform

Local space transform.

`minPoint` Vector3

Bottom left point of rect.

`maxPoint` Vector3

Top right point of rect.

Returns

(Vector3, Vector3)

World space points.

OnDestroy()

```
protected virtual void OnDestroy()
```

OnDisable()

```
protected virtual void OnDisable()
```

OnEnable()

```
protected virtual void OnEnable()
```

OnPreWillRenderCanvases()

Event that is called just before Canvas rendering happens.

```
protected virtual void OnPreWillRenderCanvases()
```

OnTransformParentChanged()

```
protected virtual void OnTransformParentChanged()
```

Resize()

Method where main resize logic should be implemented.

```
protected virtual void Resize()
```

SetRectTransformDefaults()

Sets default values for driven RectTransform.

```
protected void SetRectTransformDefaults()
```

WorldRectToLocalRect(Transform, Rect)

Converts world space rect to local space.

```
protected static Rect WorldRectToLocalRect(Transform localTransform, Rect worldRect)
```

Parameters

`localTransform` Transform

Local space transform.

`worldRect` Rect

World space rect.

Returns

Rect

Local space rect.

WorldToLocalPoints(Transform, Vector3, Vector3)

Converts world space points to local space.

```
protected static (Vector3, Vector3) WorldToLocalPoints(Transform localTransform, Vector3
minPoint, Vector3 maxPoint)
```

Parameters

`localTransform` Transform

Local space transform.

`minPoint` Vector3

Bottom left point of rect.

`maxPoint` Vector3

Top right point of rect.

Returns

(Vector3, Vector3)

Local space points.

Class UIScalerGroupSo

Namespace: SmartUIScaler

Assembly: SmartUIScaler.dll

UIScaler group scriptable object.

```
[CreateAssetMenu(fileName = "UIScalerGroup", menuName = "SmartUIScaler/UIScalerGroup")]
public class UIScalerGroupSo : ScriptableObject
```

Inheritance

object ← Object ← ScriptableObject ← UIScalerGroupSo

Properties

DefaultOptions

Default options of this group.

```
public UIScalerOptions DefaultOptions { get; }
```

Property Value

UIScalerOptions

SaveId

Identifier used to match this group in save data. Generated when the asset is created and editable in the inspector. The editor keeps it unique across the project. A group with an empty Save Id is not persisted.

```
public string SaveId { get; }
```

Property Value

string

Methods

SetContentScale(float)

Sets the content scale for this group.

```
public void SetContentScale(float contentScale)
```

Parameters

`contentScale` float

Controls the scale of the objects inside UIScalers. Recommended scale is [0.1, 1.0].

SetExpandProgress(float)

Sets the expand progress for this group.

```
public void SetExpandProgress(float expandProgress)
```

Parameters

`expandProgress` float

Controls UIScaler occupied space between reference resolution and max allowed screen space. Allowed value is [0.0, 1.0].

Class UIScalerGroups

Namespace: SmartUIScaler

Assembly: SmartUIScaler.dll

UIScaler groups manager.

```
public static class UIScalerGroups
```

Inheritance

object ← UIScalerGroups

Fields

PlayerPrefsSaveKey

Default PlayerPrefs key for saving UIScaler groups save data.

```
public const string PlayerPrefsSaveKey = "UIScalerGroupsSaveData"
```

Field Value

string

Methods

AddGroupListener(UIScalerGroupSo, Action<UIScalerOptions>, bool)

Add listener to UIScaler group options changes.

```
public static UIScalerOptions AddGroupListener(UIScalerGroupSo uiScalerGroup,
Action<UIScalerOptions> listener, bool invokeListenerOnAdd = true)
```

Parameters

`uiScalerGroup` UIScalerGroupSo

UIScaler group to listen to.

`listener` Action<UIScalerOptions>

Changes callback.

`invokeListenerOnAdd` bool

If change callback should be invoked on add.

Returns

UIScalerOptions

Current UIScaler group options.

GetSaveData()

Get current UIScaler groups save data. Use it for custom saving. Includes loaded options of groups that have not been used yet, so they are not lost. Groups with an empty Save Id are skipped. When several groups share a Save Id only the first used one is saved.

```
public static UIScalerGroupsSaveData GetSaveData()
```

Returns

UIScalerGroupsSaveData

UIScaler groups save data.

LoadFromPlayerPrefs(string)

Load UIScaler groups save data from PlayerPrefs to be applied to UIScalers.

```
public static void LoadFromPlayerPrefs(string playerPrefsSaveKey = "UIScalerGroupsSaveData")
```

Parameters

`playerPrefsSaveKey` string

PlayPrefs key.

RemoveGroupListener(UIScalerGroupSo, Action<UIScalerOptions>)

Remove UIScaler group options changes listener.

```
public static void RemoveGroupListener(UIScalerGroupSo uiScalerGroup,
Action<UIScalerOptions> listener)
```

Parameters

`uiScalerGroup` UIScalerGroupSo

UIScaler group to remove listener from.

`listener` Action<UIScalerOptions>

Changes callback.

SaveToPlayerPrefs(string)

Save current UIScaler groups save data to PlayerPrefs.

```
public static void SaveToPlayerPrefs(string playerPrefsSaveKey = "UIScalerGroupsSaveData")
```

Parameters

`playerPrefsSaveKey` string

PlayPrefs key.

SetGroupContentScale(UIScalerGroupSo, float)

Set UIScaler group content scale.

```
public static void SetGroupContentScale(UIScalerGroupSo uiScalerGroup, float contentScale)
```

Parameters

`uiScalerGroup` UIScalerGroupSo

UIScaler group to apply content scale.

`contentScale` float

Controls the scale of the objects inside UIScalers. Recommended scale is [0.1, 1.0].

SetGroupExpandProgress(UIScalerGroupSo, float)

Set UIScaler group expand progress.

```
public static void SetGroupExpandProgress(UIScalerGroupSo uiScalerGroup,
float expandProgress)
```

Parameters

`uiScalerGroup` UIScalerGroupSo

UIScaler group to apply expand progress.

`expandProgress` float

Controls UIScaler occupied space between reference resolution and max allowed screen space. Allowed value is [0.0, 1.0].

SetSaveData(UIScalerGroupsSaveData)

Set UIScaler group save data to be applied to UIScalers. Use it for custom loading. Options of groups that are not used yet are applied when they are first used. The provided save data is not modified.

```
public static void SetSaveData(UIScalerGroupsSaveData saveData)
```

Parameters

`saveData` UIScalerGroupsSaveData

UIScaler groups save data.

StoreSaveDataToPlayerPrefs(string)

Store current UIScaler groups save data to PlayerPrefs without saving.

```
public static void StoreSaveDataToPlayerPrefs(string playerPrefsSaveKey
= "UIScalerGroupsSaveData")
```

Parameters

`playerPrefsSaveKey` string

PlayPrefs key.

Class UIScalerGroupsSaveData

Namespace: SmartUIScaler

Assembly: SmartUIScaler.dll

UIScaler groups save data.

```
[Serializable]
public class UIScalerGroupsSaveData
```

Inheritance

object ← UIScalerGroupsSaveData

Fields

CurrentVersion

Format version written by GetSaveData(). Data written by plugin version 1.0 has no version and no Save Id values and is ignored on load.

```
public const int CurrentVersion = 2
```

Field Value

int

GroupOptions

List of UIScaler group options save data.

```
public List<UIScalerGroupsSaveData.UIScalerGroupOptions> GroupOptions
```

Field Value

List<UIScalerGroupsSaveData.UIScalerGroupOptions>

Version

Format version of this data. See CurrentVersion.

```
public int Version
```

Field Value

int

Class UIScalerGroupsSaveData.UIScalerGroupOptions

Namespace: SmartUIScaler

Assembly: SmartUIScaler.dll

UIScaler group options save data.

```
[Serializable]
public class UIScalerGroupsSaveData.UIScalerGroupOptions
```

Inheritance

object ← UIScalerGroupsSaveData.UIScalerGroupOptions

Fields

ContentScale

Controls the scale of the objects inside UIScalers.

```
public float ContentScale
```

Field Value

float

ExpandProgress

Controls UIScaler occupied space between reference resolution and max allowed screen space.

```
public float ExpandProgress
```

Field Value

float

SaveId

Save identifier of UIScalerGroupSo scriptable object. See SaveId.

```
public string SaveId
```

Field Value

string

Struct UIScalerOptions

Namespace: SmartUIScaler

Assembly: SmartUIScaler.dll

UIScaler options.

```
[Serializable]
public struct UIScalerOptions : IEquatable<UIScalerOptions>
```

Implements

IEquatable <UIScalerOptions>

Fields

Alignment

Controls the alignment of UIScaler inside provided screen area.

```
public UIScalerAlignment Alignment
```

Field Value

UIScalerAlignment

ContentScale

Controls the scale of the objects inside UIScalers. Recommended scale is [0.1, 1.0].

```
public float ContentScale
```

Field Value

float

ExpandProgress

Controls UIScaler occupied space between reference resolution and max allowed screen space. Allowed value is [0.0, 1.0].

```
[Range(0, 1)]
public float ExpandProgress
```

Field Value

float

ReferenceResolution

The resolution the UI layout is designed for.

```
public Vector2Int ReferenceResolution
```

Field Value

Vector2Int

UseSafeArea

Should UIScaler use only the screens safe area or whole provided screen area.

```
public bool UseSafeArea
```

Field Value

bool

Properties

Default

Default UIScaler options.

```
public static UIScalerOptions Default { get; }
```

Property Value

UIScalerOptions

Methods

Equals(UIScalerOptions)

Checks if all options are equal to the options of another instance.

```
public bool Equals(UIScalerOptions other)
```

Parameters

`other` UIScalerOptions

Options to compare with.

Returns

bool

True when every option is equal.

Equals(object)

Checks if the object is UIScaler options with all options equal to this instance.

```
public override bool Equals(object obj)
```

Parameters

`obj` object

Object to compare with.

Returns

bool

True when the object is equal UIScaler options.

GetHashCode()

Hash code combined from all options.

```
public override int GetHashCode()
```

Returns

int

Hash code.